Rainfall Prediction

Predicting next-day rainfall using historical weather data

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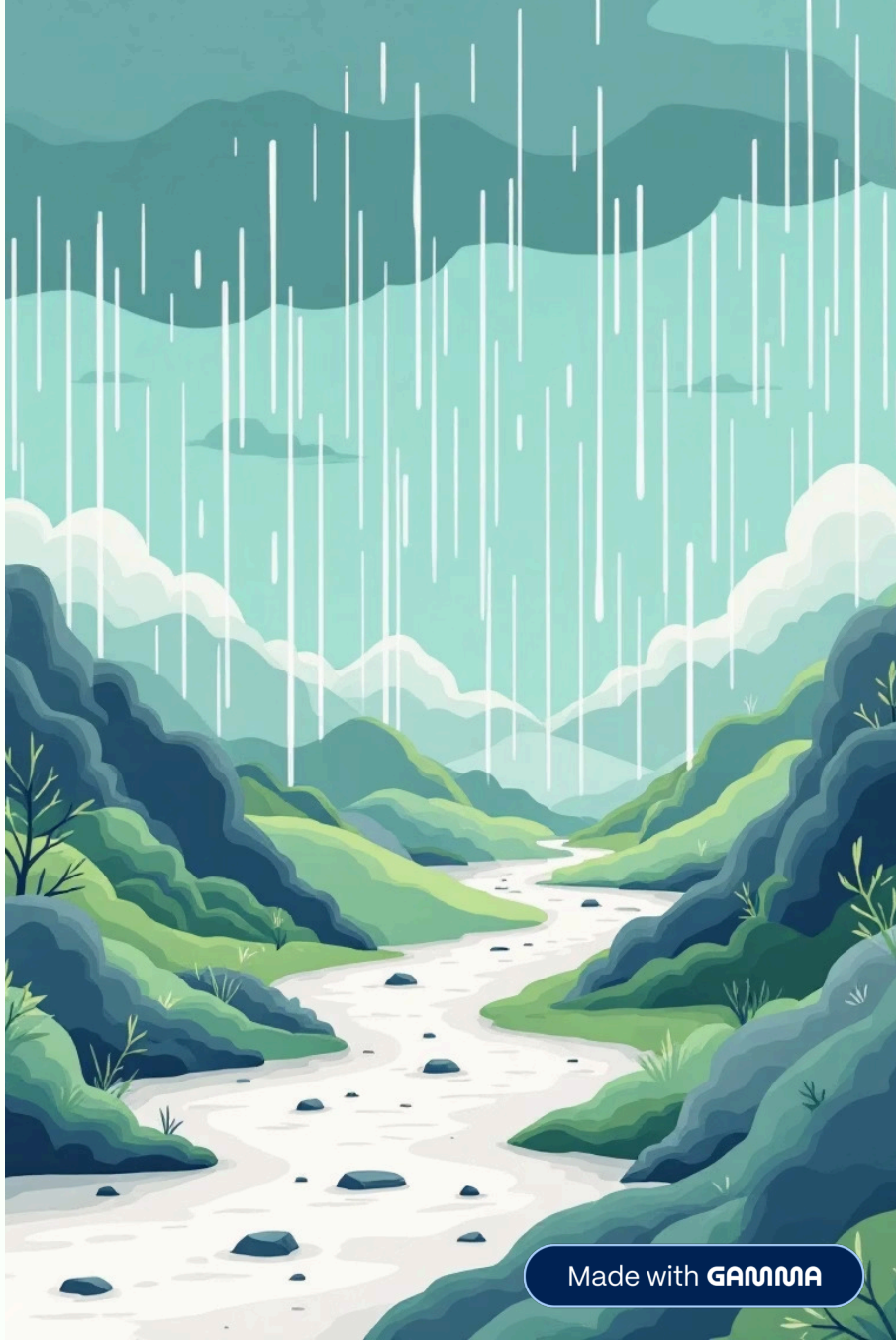

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Introduction to Rainfall Prediction

Importance

Accurate rainfall prediction is crucial for various sectors, including agriculture, transportation, and urban planning. It enables better resource management and disaster preparedness.

Challenges

Weather patterns are inherently nonlinear and complex, introducing significant uncertainty. Traditional forecasting methods often struggle with these intricate dynamics.

Our Approach

We are shifting from conventional meteorological forecasting to leveraging the power of machine learning to capture these complex relationships effectively.

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Problem Statement & Objectives

Our primary goal is to predict "RainTomorrow" (Yes/No) with high accuracy.

01

Exploratory Data Analysis (EDA)

Understand dataset characteristics, identify patterns, and detect anomalies.

02

Data Preprocessing

Clean and transform raw data into a suitable format for model training.

03

Model Training & Comparison

Develop and evaluate various machine learning models to identify the best performer.

04

Model Deployment

Integrate the chosen model into a user-friendly application for real-world predictions.

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Dataset Description: Rain in Australia

We utilised the "Rain in Australia" dataset, spanning from 2008 to 2017.

- Approximately 145,000 records
- 24 distinct weather attributes

This comprehensive dataset provides a rich source of information for training robust prediction models.

Key Features for Prediction:

1. Temperature
2. Humidity
3. Pressure
4. Wind Speed
5. Sunshine Hours
6. Rainfall Amount

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Data Preprocessing Steps

Robust preprocessing is essential for training effective machine learning models.

Handling Missing Values

Employing imputation techniques to address missing data points and maintain dataset integrity.

Encoding Categorical Features

Converting non-numeric categorical variables into a numerical format suitable for algorithms.

Scaling Numerical Variables

Normalising numerical features to prevent dominance by variables with larger scales, improving model convergence.

Feature Engineering

Creating new features like month, season, and lag features to capture temporal dependencies and enhance predictive power.

Train-Test Split

Dividing the dataset into an 80-20 ratio for training and testing, ensuring unbiased model evaluation.

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Methodology: Model Selection & Evaluation

Machine Learning Models Utilised:

- **Logistic Regression**: A linear model for binary classification.
- **Naive Bayes**: A probabilistic classifier based on Bayes' theorem.
- **Random Forest**: An ensemble method using multiple decision trees.
- **XGBoost**: A powerful gradient boosting framework.
- **CatBoost**: A gradient boosting library specifically for categorical features.

Optimisation & Evaluation:

- **Hyperparameter Tuning**: Employed Grid Search and Random Search to find optimal model parameters.
- **Evaluation Metrics**: Assessed model performance using:
  - Accuracy
  - Precision
  - Recall
  - F1-Score
  - ROC-AUC

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Key Results and Performance

Our analysis revealed superior performance from specific model types.

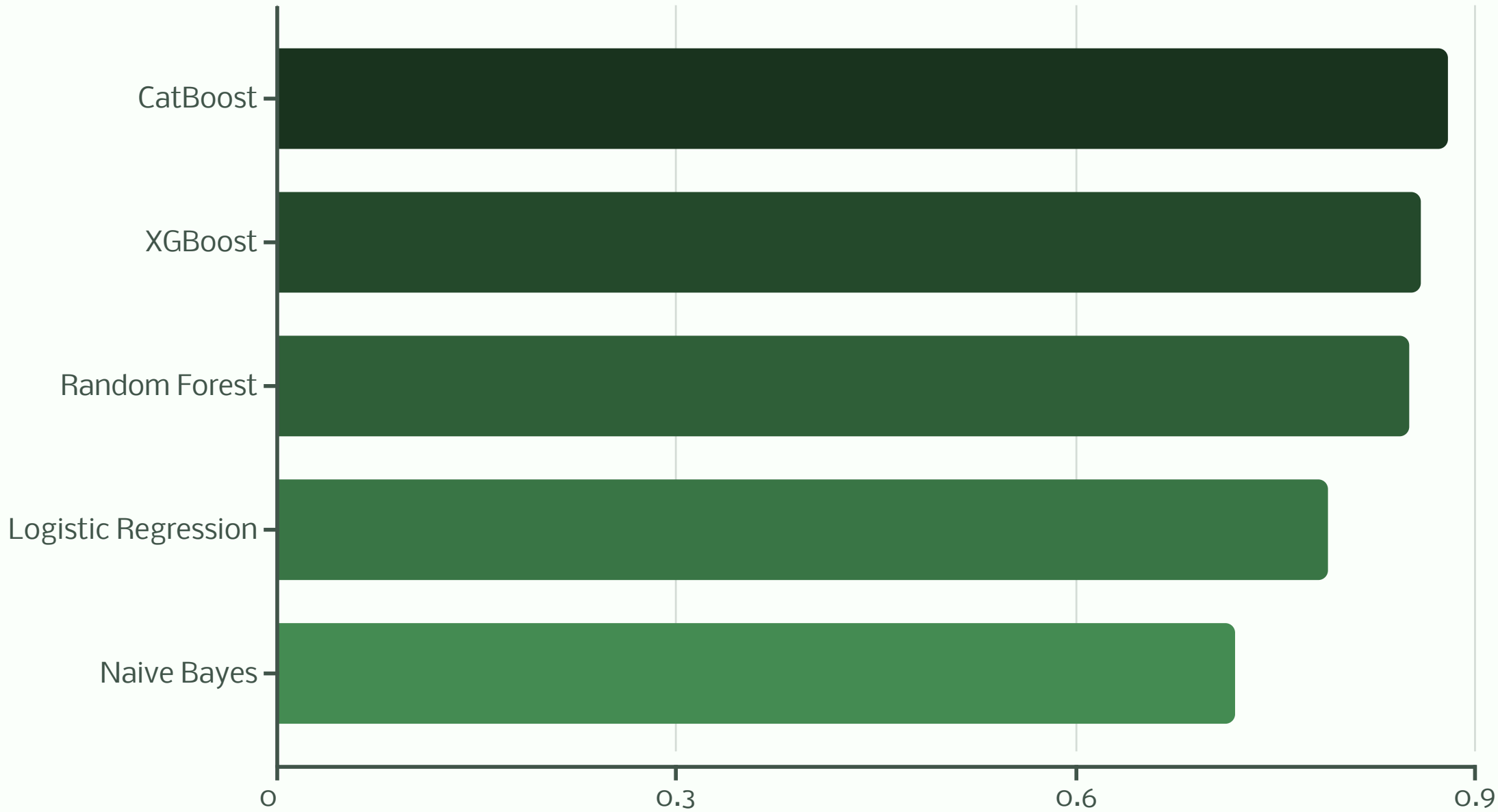
CatBoost Performance

The CatBoost model consistently delivered the best performance across all evaluation metrics, highlighting its effectiveness for this dataset.

Tree-based Models

Overall, tree-based ensemble models significantly outperformed linear models, demonstrating their ability to capture complex non-linear relationships in weather data.

Summary of Performance:



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Deployment: Web Application

To make our prediction model accessible, we developed a simple web application.

Technology Stack:

- **Flask Framework**: A lightweight Python web framework for building the application backend.
- **Joblib for Model Loading**: The trained CatBoost model is efficiently loaded using joblib for quick predictions.

User Interface:

The application features a straightforward interface where users can input relevant weather data.

It then processes this input through the deployed model and outputs whether it will be "Rainy" or "Sunny" the next day.

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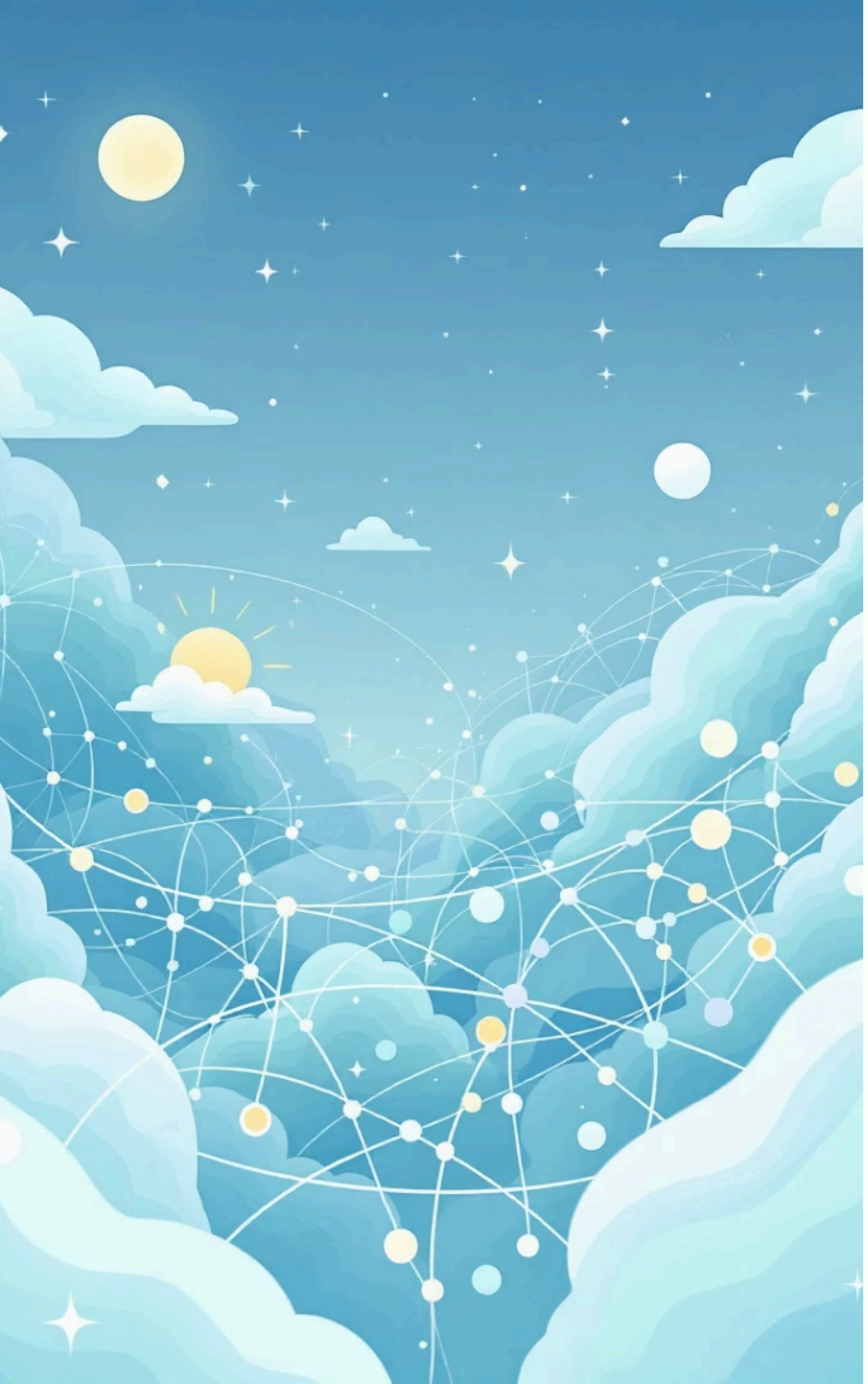

Conclusion & Future Work

Accurate Prediction Achieved

We successfully developed an accurate next-day rainfall prediction system. The models demonstrated significant capability in forecasting.

Ensemble Models Excel

Our findings confirm that ensemble learning models, particularly CatBoost, are superior for this complex prediction task.

Future Scope

Future enhancements include exploring **LSTMs/Transformers**, incorporating **satellite/radar data**, and integrating **external climate indicators** for even greater accuracy.

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